Yujie Cen

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RESEARCH FOCUS

Machine-learning interatomic potentials (MLIP) and active-learning workflows for large-scale atomistic simulation; thermal transport, phonon dynamics, and symmetry-adapted methods for low-dimensional materials.

SELECTED HIGHLIGHTS

- First-author paper in npj Computational Materials (2026): symmetry-adapted Atomistic Green's Function method for thermal transport in defect-laden quasi-1D systems.
- Lead developer of pulgon_tools — open-source Python package for line-group symmetry analysis (submitted to Computer Physics Communications).
- Trained equivariant MLIPs (Allegro, NEP) for CNT and TMD nanotubes via active learning; ≤1.2 meV/atom energy RMSE on DFT reference.
- Supervisor (G. K. H. Madsen) collaborates directly with G. Kresse (Univ. Vienna, VASP) — warm-introduction channel.

TECHNICAL SKILLS

ML for atomistic simulation	Allegro, MACE, NequIP, NEP (GPUMD)
Languages & frameworks	Python, MATLAB, PyTorch, JAX, C++ (basic), Bash
Atomistic codes	VASP, GPUMD, LAMMPS, ASE, Phonopy, pymatgen
Thermal & transport methods	Green-Kubo, GKMA, HNEMA, SED · Atomistic Green's Function · BTE
Infrastructure	SLURM, Git, Linux

EDUCATION

PhD in Materials Chemistry · TU Wien, Austria · Jul 2023 – Present (expected late 2026)
Supervisor: Prof. Georg K. H. Madsen (Institute of Materials Chemistry, E165, TU Wien)
Co-supervisor: Dr. Jesús Carrete Montaña (INMA, CSIC–Univ. Zaragoza, Spain)

MS in Theoretical Physics · South China University of Technology · Sept 2020 – Jul 2023

BS in Physics · South China University of Technology · Sept 2016 – Jul 2020

RESEARCH EXPERIENCE

PhD Candidate · TU Wien · Jul 2023 – Present

Symmetry-adapted MLIP-driven thermal transport in quasi-1D systems *(in preparation)*
Extending Green-Kubo Modal Analysis (GKMA) and HNEMA with line-group projector operators so every phonon mode carries a well-defined irrep label. Trained a NEP foundation-model fine-tune for (10,0) CNT via two-round active learning (VASP/PBEsol-D3 reference): 1.04 meV/atom energy RMSE, 120 meV/Å force RMSE; production GPUMD trajectories (10–40 ns, 200–400 K) confirm irrep-summed $\kappa = 2538$ W m^{-1} K^{-1} is invariant across basis constructions, demonstrating physically interpretable symmetry-resolved thermal conductivity.

Ab-initio heat transport in defect-laden quasi-1D systems *(npj Computational Materials, 2026 · first author)*
Discovered that strong symmetry breaking can *enhance* thermal conductance in WS_2/MoS_2 multi-wall nanotubes by relaxing line-group selection rules and opening previously forbidden phonon channels — a counterintuitive result established with a symmetry-adapted, mode-resolved Atomistic Green's Function (AGF) framework that fuses representation theory with line-group projector techniques to label phonon transmission by irrep. An Allegro-based equivariant MLIP supplied interatomic force constants at near-DFT accuracy, and Green-Kubo MD confirmed the conclusion at finite temperature.

pulgon_tools *(Computer Physics Communications, submitted · first author, lead developer)*
Open-source Python/CLI package for quasi-1D systems: chiral-rolling and symmetry-based structure generation, line-group detection (axial point group + generalized translation group), q-point–resolved character tables and

irrep matrices, and convex-optimisation–based symmetrisation of force constants (enforcing translational and Born–Huang rotational sum rules). Apache-2.0 licensed with CLI entry points, test suite, and PyPI release. github.com/pulgon-project/pulgon_tools

Symmetry-guided high-throughput search for alloy ground states *(Physical Review Materials, 2022 · first author)*
Partitioned the cluster-expansion correlation-function space by symmetry (Wyckoff-position index) so the convex hull can be solved subspace-by-subspace, drastically cutting the candidate set for first-principles screening. Applied to Ni–Pt, Ag–Pd, Os–Ru, Ir–Ru, and Mo–Ru, recovering known ground states and predicting several new lower-energy configurations.

PUBLICATIONS

First-author papers

1. **Cen, Y.**, Wieser, S., Madsen, G. K. H., & Carrete, J. (2026). Ab-initio heat transport in defect-laden quasi-1D systems from a symmetry-adapted perspective. *npj Computational Materials*, 12(1), 19. doi.org/10.1038/s41524-025-01866-1
2. **Cen, Y.**, Wieser, S., Madsen, G. K. H., & Carrete, J. (2025). Pulgon-tools: a toolkit for analysing and harnessing symmetries in quasi-1D systems. *Computer Physics Communications* (submitted). arXiv:2603.28435
3. **Cen, Y.**, He, C. C., Qiu, S. B., Zhao, Y. J., & Yang, X. B. (2022). Determining ground states of alloys by a symmetry-based classification. *Physical Review Materials*, 6(5), L050801. doi.org/10.1103/PhysRevMaterials.6.L050801

Co-authored papers

4. Wieser, S., **Cen, Y.**, Madsen, G. K. H., & Carrete, J. (2025). Accelerating first-principles molecular-dynamics thermal conductivity calculations for complex systems. *Journal of Chemical Theory and Computation*. doi.org/10.1021/acs.jctc.5c01525
5. Zhu, C., Yan, X., **Cen, Y.**, He, C. C., Zhao, Y. J., & Yang, X. B. (2025). Accelerating the identification of stable configurations in mixed-anion perovskite materials. *Computational Materials Science*, 248, 113564. doi.org/10.1016/j.commatsci.2025.113564
6. Li, D. Z., **Cen, Y.**, Wang, X., & Yang, X. B. (2024). Residual entropy of ice: a study based on transfer matrices. *Physical Review B*, 110(19), 195414. doi.org/10.1103/PhysRevB.110.195414
7. Xiong, J., Peng, Y. H., Lin, J. Y., **Cen, Y.**, Yang, X. B., & Zhao, Y. J. (2023). High concentration intrinsic defects in $MnSb_2Te_4$. *Materials*, 16(15), 5496. doi.org/10.3390/ma16155496
8. Lin, J. Y., Chen, Z. J., Li, G. L., Zeng, J., **Cen, Y.**, et al. (2022). Theoretical study of magnetic phase transition in $La_{2/3}M_{1/3}MnO_3$ (M = Ca, Sr) membranes. *Physics Letters A*, 432, 128010. doi.org/10.1016/j.physleta.2022.128010